

Roadmap: Stage-Wise Attribution and Explainability Across IR Pipelines

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Course: CS4.406 — Information Retrieval & Extraction

Domain: Explainable AI + Information Retrieval

Research Questions

The project is structured around three progressively harder questions, forming a natural fallback hierarchy — the project produces a valid deliverable even if only RQ1 succeeds.

RQ1 — Attribution. Can we decompose a multi-stage retrieval pipeline's final ranking into per-stage contributions for each document?

RQ2 — Comparative Behavior. Do those attribution patterns differ systematically when the first-stage retriever is swapped (e.g., BM25 vs. Contriever vs. ColBERT)?

RQ3 — Faithfulness. Do the computed attributions correspond to actual causal influence — does removing the stage identified as most important actually change the outcome?

Pipeline Architecture

Query



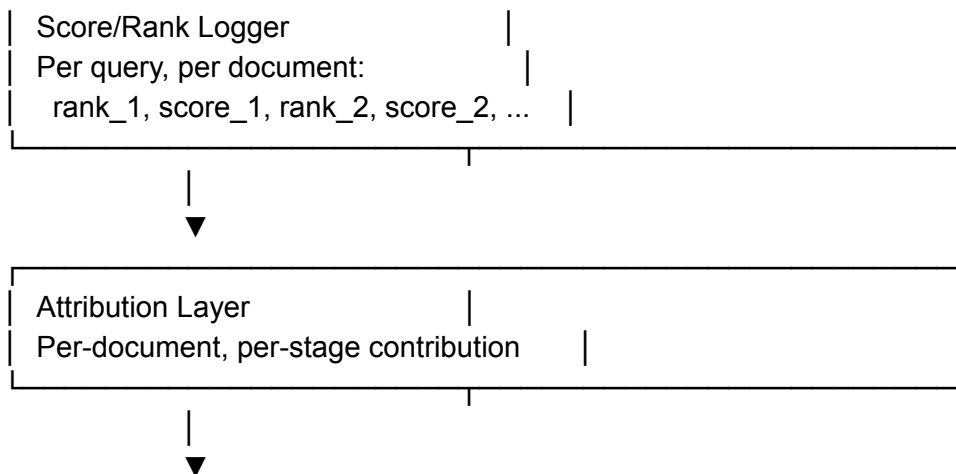
Stage 1: First-Stage Retriever (swappable)
Input: query + full corpus
Output: top-K candidates + $\text{score_1}(q, d)$



Stage 2: Reranker (swappable)
Input: query + K candidates from Stage 1
Output: reranked list + $\text{score_2}(q, d)$



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Final top-K + explanations
 Every stage exposes raw scores and resulting ranks for every document passing through it.
 This full trajectory is the data the attribution layer operates on.

Design Choices

First-Stage Retrievers

Retriever	Type	Key Property	Implementation
BM25	Sparse lexical	Exact term matching, IDF weighting	Pyserini (wraps Lucene)
Contriever	Dense bi-encoder	Unsupervised contrastive pre-training	HuggingFace: facebook/contriever
ColBERT	Late-interaction	Token-level matching, richer than bi-encoder	ColBERTv2 via Stanford repo
SPLADE	Learned sparse	Learned term expansion into sparse vectors	naver/splade-v3
Hybrid (BM25 + Dense)	Fusion	Reciprocal rank fusion	Custom: run both, fuse with RRF

Minimum: BM25 + Contriever (one sparse, one dense).

Recommended: Add ColBERT as a third paradigm for richer comparison.

Extension: SPLADE and/or Hybrid if time permits.

Rerankers

Reranker	Type	Implementation
MiniLM cross-encoder	Cross-encoder	cross-encoder/ms-marco-MiniLM-L-12-v2
MonoT5	Seq2seq reranker	castorini/monot5-base-msmarco
LLM reranker	Listwise LLM	Llama-3-8B with RankGPT-style prompting
No reranker	Control baseline	Stage 1 ranking passed through unchanged

Minimum: MiniLM cross-encoder (fast, well-understood) + no-reranker baseline.

Extension: MonoT5 or LLM reranker to test whether attribution patterns change with reranker strength.

Datasets

Dataset	Corpus	Queries	Relevance Judgments	Why
MS MARCO Passage	8.8M passages	6,980 dev	Sparse binary	Standard benchmark, largest retriever ecosystem
TREC-DL 2019	Same corpus	43 queries	Dense graded (NIST)	Gold-standard graded relevance for nDCG
TREC-DL 2020	Same corpus	54 queries	Dense graded (NIST)	Second evaluation set, same corpus
BEIR subsets	Various	Various	Various	Cross-domain generalization (extension)

Recommendation: MS MARCO + TREC-DL 2019/2020 — same corpus (index once), and TREC-DL provides graded relevance judgments needed for meaningful nDCG computation.

Attribution Methods

Approach A: Rank-Delta Attribution (Baseline)

For each document D in the final top- K , track rank change at each stage:

$$\Delta_{\text{stage}_i}(D) = \text{rank}_{\{i-1\}}(D) - \text{rank}_i(D)$$

Positive $\Delta \rightarrow$ stage promoted D (helped it)

Negative $\Delta \rightarrow$ stage demoted D (hurt it)

Normalize across stages for fractional contributions.

Pros: simple, requires no model access, works with any pipeline.

Cons: rank changes are relative (promoting one document necessarily demotes others); doesn't distinguish inclusion from ordering.

Approach B: Inclusion-vs-Ordering Decomposition (Main Method)

Decompose each stage's contribution into two distinct effects:

Stage 1 contribution:

- Inclusion effect: did D make the top- K candidate cut? (binary gate)
- Ordering effect: given D was included, what initial rank did it get? (continuous)

Stage 2 contribution:

- Reordering effect: how much did the reranker move D within the candidate set? (continuous)

Separates the "gatekeeper" function of Stage 1 (setting the candidate boundary) from its "ranking" function. More nuanced than raw rank-delta.

Approach C: Shapley-Inspired Stage Attribution (Extension)

Treat each pipeline stage as a "player" in a cooperative game. Value function = some ranking quality metric ($\text{nDCG}@10$, or document-specific "is D in the top- K ?"). Compute each stage's Shapley value: average marginal contribution across all subsets of stages.

For 2 stages: 4 subsets. For 3 stages: 8 subsets. Computationally trivial at this scale.

Open design question: what does it mean to "run Stage 2 without Stage 1"? Options include feeding the reranker the full corpus, a random sample, or defining the value of {Stage 2 alone} as zero. Each choice has implications to be resolved during implementation.

Plan: Implement A as baseline, B as main method, C if time permits. The comparison between methods is itself a contribution.

Faithfulness Evaluation

Intervention Types

Intervention	Procedure	What It Tests
Stage removal	Remove Stage 2, use Stage 1 ranking as final	Does removing the "important" stage cause predicted documents to drop?
Score perturbation	Add calibrated noise to one stage's scores, re-run downstream	Does perturbing the attributed-as-important stage cause larger rank changes?
Candidate set perturbation	Remove D from Stage 1 output, re-run Stage 2	Tests Stage 1's gatekeeper function for specific documents
Score replacement	Replace Stage 1 scores with uniform random, keep candidate set	Isolates ordering contribution from inclusion contribution
Stage swap	Replace BM25 with Contriever as Stage 1, keep same reranker	Do "Stage 1 driven" docs change rank more than "Stage 2 driven" docs?

Faithfulness Metrics

Adapted from DeYoung et al. (2020) ERASER benchmark:

Comprehensiveness. If attribution says Stage X is most important for document D, removing Stage X should cause D to drop from top-K. Comprehensiveness = fraction of such predictions that are correct.

Sufficiency. If attribution says Stage X is most important for document D, keeping only Stage X should be enough for D to remain in top-K. Sufficiency = fraction of such predictions that hold.

Rank-Sensitivity Correlation. Across all documents, correlate `attributed_importance(stage, doc)` with `actual_rank_change_upon_removal(stage, doc)`. High correlation = faithful attribution.

Experiments

Core Experiments (Must Complete)

#	Experiment	Independent Variable	Dependent Variable	Question
E1	Attribution distribution	Retrieval paradigm	Fraction of Stage-1-driven vs. Stage-2-driven docs in final top-K	Do paradigms produce different attribution patterns?
E2	Reranker value-add	Retrieval paradigm	nDCG@10 improvement from adding reranker	Is the reranker equally valuable across paradigms?
E3	Attribution faithfulness	Attribution method	Comprehensiveness, sufficiency, rank-sensitivity correlation	Are the computed attributions causally correct?
E4	Method comparison	Attribution method (rank-delta vs. inclusion-ordering)	Agreement between methods, faithfulness scores	Which attribution approach is most reliable?
E5	Paradigm agreement	Retrieval paradigm	Overlap in final top-K, attribution of shared vs. unique docs	Why do different paradigms agree or disagree on final results?

Extension Experiments (If Time Permits)

#	Experiment	Independent Variable	Dependent Variable	Question
E6	Query-type slicing	Query category (entity, semantic, rare-term, ambiguous)	Attribution distribution per category	Do entity queries behave differently from semantic queries?
E7	Reranker strength	Reranker type (MiniLM vs. MonoT5 vs. LLM)	Attribution distribution, faithfulness	Does a stronger reranker change the attribution pattern?
E8	Candidate set size	Top-K cutoff at Stage 1 (100, 500, 1000)	Attribution distribution, final nDCG@10	Does the gatekeeper boundary affect which stage "matters"?

Reach Experiments

#	Experiment	Independent Variable	Dependent Variable	Question
E9	Score perturbation sensitivity	Noise magnitude added to Stage 1 or Stage 2	Rank displacement in final top-K	Continuous measure of causal influence per stage
E10	Cross-domain generalization	Dataset (MS MARCO vs. BEIR subset)	Attribution patterns	Do attribution dynamics generalize across domains?

Minimum Viable Project

If time runs short, the project still produces a valid result with:

MVP: 1 retriever + 1 reranker + 1 attribution method + faithfulness eval

(Phase 1 + Phase 2 + Phase 3)

Extension 1: Multiple retrieval paradigms + cross-paradigm analysis
(+ Phase 4)

Extension 2: Query-type slicing + second reranker
(+ E6, E7)

Extension 3: Interactive debugger
(+ Phase 5)

Each extension adds value independently. The project degrades gracefully under time pressure.

Team Division

Member	Primary Responsibility	Phases
Member 1	Retrieval infrastructure: retriever wrappers, reranker wrappers, indexing, pipeline orchestrator, profiling	Phase 1, Phase 4 (new retrievers)
Member 2	Attribution methods: rank-delta, inclusion-ordering, Shapley; faithfulness evaluation; intervention tests	Phase 2, Phase 3
Member 3	Experimental analysis: query-type categorization, cross-paradigm comparison, statistical analysis, interactive debugger	Phase 4, Phase 5

All members contribute to: Phase 0 (literature review), Phase 6 (report), and research design decisions throughout.

Resources

Resource	Requirement	Availability
Compute	BM25 indexing (CPU); bi-encoder inference (GPU)	Free-tier GPU (Colab T4, Kaggle P100) sufficient

Resource	Requirement	Availability
	helpful); cross-encoder reranking (GPU helpful, CPU feasible)	
Storage	MS MARCO passage index (~15–20 GB with all retriever indexes)	Colab/Kaggle disk or local machine
Data	MS MARCO Passage, TREC-DL 2019/2020 — publicly available	Yes
Model weights	Contriever, ColBERT, MiniLM cross-encoder, MonoT5 — all on HuggingFace	Yes
Retrieval toolkits	Pyserini (BM25), sentence-transformers (bi-encoder), HuggingFace (cross-encoder)	Yes, open-source
Skills	Python, PyTorch, IR evaluation (nDCG, MRR), Shapley values	Covered by course + ML background
API costs	None — entirely open-weight models	Zero cost

Key Risks

Risk	Likelihood	Impact	Mitigation
Attribution method is too simple to be considered a contribution	Medium	High	The <i>evaluation framework</i> (faithfulness testing for pipeline-level attribution) is the contribution even if the method itself is simple; the

Risk	Likelihood	Impact	Mitigation
			comparison across paradigms adds novelty
ColBERT/SPLADE indexing is slow or complex to set up	Medium	Medium	Start with BM25 + Contriever only; add ColBERT in Phase 4 only after core results are solid
Faithfulness interventions are ambiguous to interpret (e.g., removing Stage 1 changes everything trivially)	Medium	High	Use multiple intervention types (not just removal); score perturbation and candidate-set perturbation are more surgical
Existing work already handles multi-stage attribution (discovered in Phase 0)	Low	High	Phase 0 explicitly checks this; if found, pivot to extending/evaluating that work rather than proposing from scratch
Interactive debugger takes too long to build	Medium	Low	Debugger is Extension 3 — project is complete without it; use Streamlit for fast MVP

Papers to Survey in Depth

Paper	Why
Rank-LIME (Chowdhury et al., ICTIR 2023)	Closest existing work on ranking explainability; confirm it doesn't handle multi-stage

Paper	Why
RankingSHAP (Heuss, de Rijke & Anand, SIGIR 2025)	Shapley-based ranking attribution; understand axioms for potential adaptation to stage level
ERASER (DeYoung et al., 2020)	Comprehensiveness/sufficiency metrics — the basis of our faithfulness evaluation
Câmara & Hauff (ECIR 2020)	Probing BERT rankers with retrieval heuristics — useful for understanding what rerankers encode
Anand et al. (2022) survey	Full ExIR taxonomy; confirm where stage-level attribution sits or doesn't
Anand et al. (SIGIR 2023 tutorial)	Tutorial on Explainable IR; identifies "retrieval cascades" as open problem
Saha, Majumdar & Mitra (ACM Computing Surveys, 2026)	Most recent survey; confirm pipeline-level attribution remains unexplored
BEIR (Thakur et al., NeurIPS 2021)	Benchmark methodology for cross-paradigm comparison